

Robert Chuang

Senior Mobile App Solution Architect

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Professional Summary

Senior Mobile Solution Architect with **16 years** of expertise in iOS and cross-platform development, specializing in **AI integration, performance optimization, and complex system architecture design**. Proven track record of migrating server-side AI recognition systems to mobile platforms using **C++ FFI architecture**, achieving **90% performance improvement** (5s → 0.5s response time). Led enterprise-grade applications serving government agencies and Fortune 500 companies. Expert in **Swift, Objective-C, Flutter, C++** with deep experience in **large-scale modular architecture, multi-target management, and cross-platform integration**. Strong advocate for **AI-powered development workflows** and rapid technology adoption.

Core Competencies & Technologies

Mobile Platforms	iOS (Swift, Objective-C), Flutter (Dart), Android (Java)
AI & Performance	On-device AI, C++ FFI, ONNX Runtime, OpenCV, Face Recognition, Performance Optimization
Architecture & Design	MVVM, BLoC, Clean Architecture, Modular Design, Cross-platform Integration
Backend Integration	PHP Laravel, RESTful API, Firebase (FCM, Auth), WebSocket, Push Notifications (APNs, FCM)
State Management	Provider, BLoC, Riverpod, Offline-first Architecture, Hive Local Storage

DevOps & Tools	Git, CocoaPods, CI/CD, Multi-target Management, Static/Dynamic Library Integration
Specialized Features	Beacon Indoor Positioning, i18n/Localization, QR Code Integration, Offline Data Sync
AI-Powered Development	AI Coding Assistants, Rapid Prototyping, Legacy Code Analysis, Cross-domain Learning

Professional Experience

Senior iOS/Flutter Engineer & AI Integration Lead

EZPlus Technology Co., Ltd. | Application Software Development

November 2024 ~ Present

- AI System Migration Leadership:** Spearheaded migration of server-side facial recognition system to mobile platform using **Flutter FFI + C++ architecture**, achieving **90% performance improvement** from 5 seconds to 500ms response time

Impact: 90% performance boost • Apple Store compliance • Reusable Flutter Plugin SDK • Zero-network AI capability
 - Enterprise App Development:** Independently developed employee attendance management system with **face recognition, Beacon indoor positioning, offline-first architecture**, and anti-fraud mechanisms including time tampering detection
 - Large-scale Architecture:** Collaborated on enterprise facility management platform with **120K lines of code, 531 files**, implementing **BLoC + Riverpod hybrid architecture** with multi-language support (Traditional Chinese, Simplified Chinese, English)
 - Cross-domain Technical Rescue:** Rapidly learned **PHP Laravel** from zero to production-ready in crisis situation, successfully restored enterprise push notification system for multiple apps while maintaining original app development timeline
 - Technology Stack:** Flutter, iOS (Swift/Objective-C), C++, PHP Laravel, Firebase, RESTful API, Performance Engineering, System Architecture Design
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Senior Flutter Engineer & Team Lead

FUHOM Technology Co., Ltd. | Communication Equipment Industry | Team Size: 5-8 Engineers

October 2023 ~ August 2024 (11 months)

- **Internal Flutter App Development:** Led development of company's internal Flutter applications integrating with hardware devices (DVR, IP cameras) through custom API interfaces
 - **Native-Flutter Bridge:** Architected bidirectional communication between native iOS/Android modules and Flutter components for seamless integration
 - **Team Leadership:** Managed 5-8 engineer team, responsible for technical decision-making, task allocation, and delivery timeline management
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Senior iOS Engineer & iOS Platform Lead

SkyEye Satellite Technology Co., Ltd. | Software & Internet Services

October 2015 ~ July 2023 (7 years 10 months)

- **Cross-functional Team Leadership:** Led App team coordination with PM, Backend, UI, and Android teams to ensure on-time project delivery and quality standards
- **iOS Platform Ownership:** Served as primary iOS platform owner, responsible for all core module development, architecture design, and maintenance strategies
- **Engineering Excellence & Mentorship:** Trained and mentored junior engineers, established coding standards and development workflows, elevated overall team technical capabilities
- **API Architecture Modernization:** Led migration from SOAP (.asmx) to RESTful API architecture, implemented platform unification and multi-tenant support
- **Complex Multi-target Management:** Designed and maintained 10+ customized brand versions with multi-target architecture, ensuring stability and scalability
- **Key Project Deliveries:**
 - **Government Contracts:** Waste management service apps for Taichung, Chiayi, Yunlin, and Kaohsiung cities
 - **Enterprise Projects:** Apps for Yamato Transport (Black Cat), Chi Mei Corporation, Global Express
 - **Large-scale Events:** Dajia Mazu Pilgrimage app supporting tens of thousands of concurrent users

- **National Park Series:** Shei-Pa National Park iOS app, unified platform supporting Shei-Pa, Taroko, and Yushan National Parks

Key Projects & Technical Achievements

Server-to-Mobile AI Migration Project

Leading Technical Innovation | Difficulty: ★★★

- **Challenge:** Migrate Python-based server AI recognition system to Flutter mobile platform with zero-network capability
- **Solution:** Designed three-tier architecture: C++ ONNX Runtime → Flutter Plugin → Mobile App
- **Technical Breakthrough:** Overcame Apple's Nested Dynamic Frameworks limitation by converting to static library compilation

Results: 90% performance improvement (5s → 500ms) • Apple Store compliance • SDK reusability • Production deployment across 3 product lines

Enterprise Attendance Management System

Full-stack Mobile Development | Difficulty: ★★

- **Offline-first Architecture:** Implemented network-aware attendance system with intelligent mode switching, ensuring functionality under any network condition
- **Anti-fraud Technology:** Developed multi-layer security including time tampering detection, offline data integrity verification, and face recognition integration
- **Cross-platform Push Integration:** Designed Method Channel communication to bridge existing APNS infrastructure with Firebase FCM

Technology Stack: Flutter • MVVM • Provider • Hive • Beacon • FFI Integration

Enterprise Facility Management Platform

Large-scale Collaborative Development | Difficulty: ★

- **Large-scale Architecture Management:** Collaborated on complex project with **120K lines of code, 531 files**, implementing BLoC + Riverpod hybrid state management
- **Internationalization Excellence:** Implemented comprehensive multi-language architecture supporting English, Traditional Chinese, Simplified Chinese with dynamic language switching
- **Enterprise-grade Integration:** Developed modular system covering meeting rooms, equipment maintenance, laboratory, catering, and package management

Scale: 7-8 engineer team • Multi-factory support • Complex permission management • Real-time notifications



Comprehensive Technical Portfolio

<https://pub-7ffd7755217c4c84be4b81e3cddd2ad0.r2.dev/index.html>

Interactive showcase featuring 9 technical projects with video demonstrations, architecture diagrams, and detailed technical challenge explanations

Education

Master's Degree in Construction & Project Management

Feng Chia University, Taiwan | 2019 - 2021

Bachelor's Degree in Computer Science

I-Shou University, Taiwan | 2003 - 2008

Additional Qualifications

- **AI-Powered Development:** Extensive experience with AI coding assistants and automated development workflows
- **Crisis Management:** Proven ability to rapidly acquire new technologies (PHP Laravel) under emergency conditions

- **Multi-target iOS Architecture:** Expert in managing 10+ customized brand versions with complex dependency management
- **Government & Enterprise Projects:** Successfully delivered apps for national parks, municipal services, and Fortune 500 companies
- **Performance Engineering:** Deep expertise in memory optimization, startup time reduction, and cross-platform performance tuning
- **Language Skills:** Native Chinese, Professional English, Open to international collaboration